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STA301 QUESTIONS FILE

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Current solved paper (2021) Sta301

Question:01

Find $\hat{p}_c$ by using information.

Solution:

We are given that

$\hat{p}_1 = 0.31 \quad , \quad \hat{p}_2 = 0.24$

$n_1 = 100 \quad , \quad n_2 = 100$

$\hat{p}_c = ?$

We know that

$$\hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}$$

putting values,

$$\hat{p}_c = \frac{100(0.31) + 100(0.24)}{100 + 100}$$

$$\hat{p}_c = \frac{31 + 24}{200} = \frac{55}{200}$$

$$\hat{p}_c = 0.275$$

Question:02

If E(x)=0.25 then find E(2x)

Solution:

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Given that ,

$E(x) = 0.25$

We have to find ,

$E(2x) = ?$

$E(2x) = 2E(x)$

$E(2x) = 2(0.25)$

$E(2x) = 0.5$

Question:03.

Given $N=10, n=4$ & $k=5$.then find $E(x)$ and $Var(x)$

Solution:-

We know that the given data is in hypergeometric distribution.

So,

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$$E(x)=\frac{nk}{N}$$

$$E(x)=\frac{4(5)}{10}=\frac{20}{10}$$

$$\Rightarrow E(x)=2$$

and,

$$Var(x)=\frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

$$Var(x)=\frac{4(5)}{10}\left(1-\frac{5}{10}\right)\left(\frac{10-4}{10-1}\right)$$

$$Var(x)=2\left(\frac{5}{10}\right)\left(\frac{6}{9}\right)$$

$$Var(x)=\frac{6}{9}$$

$$\Rightarrow Var(x)=0.6667$$

Question:04.

Write the names of the measures of dispersion that are not based on all values?

Answer:

1) **Range.**
2) **Quartile deviation.**

Question:05.

What is a poison distribution?

Answer:

A poison distribution is also known as rare events distribution.

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Poison distribution is limiting form of binomial distribution under conditions.

These conditions are:

- **'n' trials are efficiently large.**
- **'p' very small →0 (approaches to 0).**
- **0.05 ≤ np ≤ 20**

The probability function of poison distribution is

$$P(X=x)=\frac{e^{-\mu}\mu^{x}}{x(\text{factorial})}$$

$$x=0,1,2,3,\ldots\ldots,\infty$$

<u>Question:06.</u>

Write the range, parameters and formulas for hypergeometric distribution.

<u>Answer:</u>

<u>Range:</u>

The range of hypergeometric distribution is 0,1,2,⋯..n ≤ N.

<u>Parameters:</u>

The hypergeometric distribution has three Parameters.

N, k, n.

<u>Formulas:</u>

The distribution function is

$$P(X=x)=\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$$

$$Mean=\frac{nk}{N}$$

$$Variance=\frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Question:07.

Write the range, parameters and formulas for binomial distribution.

Answer:

Range:

The range of binomial distribution is 0,1,2,…..,n

Parameters:

The binomial distribution have only 2 Parameters.

n, p.

Formulas:

The distribution function is

$$P(X=x)=\binom{n}{x}p^{n}.q^{n-x}$$

$$Mean=n.p$$

$$Variance=n.p.q$$

Question:08

A continuous random variable X that can assume b/t x=1 and x=4 has a density function given by

F(x)=1/3.

a) ***Show that area under curve is equal to 1.***
b) ***Find p(1.5<x<3).***

Solution:

a) ***We have to show that***

$$\int_{-\infty}^{\infty} f(\mathrm{x})\,\mathrm{dx} = 1$$

by putting values,

$$\int_1^4 \frac{1}{3}\,\mathrm{dx} = 1 \quad , \quad \frac{1}{3}\int_1^4 \mathrm{dx} = 1$$

$$\frac{1}{3}|x|_1^4 = 1 \quad , \quad \frac{1}{3}(4-1) = 1$$

$$\frac{1}{3}(3) = 1 \quad \Rightarrow 1 = 1$$

proved.

b) ***We have to show that p(1.5<x<3).***

$$p(1.5 < x < 3) = \int_{1.5}^{3} \frac{1}{3}\,\mathrm{dx}$$

$$p(1.5 < x < 3) = \frac{1}{3}\int_{1.5}^{3} \mathrm{dx}$$

$$= \frac{1}{3}|x|_{1.5}^{3} \quad , \quad = \frac{1}{3}(3-1.5)$$

$$= \frac{1}{3}(1.5)$$

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$$\Rightarrow p(1.5 < x < 3) = 0.5$$

Proved.

Question:09.

Critical region for following hypothesis,

$$H_2 : \sigma^2 = 25 \ , \ H_1 : \sigma^2 \neq 25$$
$$where,$$
$$\alpha = 0.05 \ , \ n = 25$$

Solution:

Critical region

$$|Z| < Z_{1-\alpha/2}$$

$$\Rightarrow \alpha / 2 = 0.025$$
$$\Rightarrow 1 - \alpha / 2 = 0.975$$

$$Z_{1-\alpha/2} = 1.96$$
$$\Rightarrow |Z| < 1.96$$

Question:10.

If E(x)=10 , E(y)=9 then find $E\left(\frac{5}{10}x + \frac{2}{6}y\right)$.

Solution:

We have to show that

$$E\left(\frac{5}{10}x + \frac{2}{6}y\right)$$

Now,

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$$= \frac{5}{10}E(\text{x}) + \frac{2}{6}E(\text{y})$$

by putting values,

$$= \frac{5}{10}(10) + \frac{2}{6}(9)$$

simplyfying,

$$= 5 + 3, = 8$$

$$\Rightarrow E\left(\frac{5}{10}x + \frac{2}{6}y\right) = 8$$

Question:11

In a normal distribution how much data that fall in interval $\mu \pm 3\sigma$ ***.***

Solution:

In normal distribution 99.73% data fall in the $\mu \pm 3\sigma$ **interval .**

(we can also be written as $\mu \pm \sigma \; and \; \mu \pm 2\sigma$ **and** $\mu \pm \sigma = 68.27\%$, $\mu \pm 2\sigma = 95.45\%$**).**

Question:12

A random sample of size 12 is drawn from normal population with mean = $\mu_0 = 10.9$ ***and variance*** σ^2 ***.***

If $S = 8$, $\bar{x} = 6.4$ ***.using information complete t-statistic.***

Solution:

As we know that

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$$t = \frac{\bar{x} - \mu_0}{S/\sqrt{n}}$$

$$= \frac{6.4 - 10.9}{8/\sqrt{12}}$$

$$= \frac{-4.5}{2.309}, = -1.948$$

Question:13

Assume that X has a poison distribution with 2. Calculate P(X=2).

Solution:

As we know,

The distribution function of poison distribution is

$$P(X = x) = \frac{e^{-\mu}\,\mu^{x}}{x(\text{fact})}\text{.........}(1)$$

Given,

Variance =2,

In poison distribution,

Variance=mean=μ=2

$$\textit{putting values in}\,(1)$$

$$P(X = x) = \frac{e^{-2}\,(2)^{2}}{2(\text{fact})}$$

$$P(X = x) = \frac{(0.1353)\,4}{2}$$

$$P(X = x) = 0.2706$$

Question:14

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Let T_1 and T_2 are two unbiased estimators, variance is
$Var(T_1) = \frac{11}{9}\sigma^2$ *and* $Var(T_2) = \frac{13}{9}\sigma^2$ ***.Which estimator is more efficient and why?***

Solution:

We know that,

$$E_f = \frac{Var(T_2)}{Var(T_1)} = \frac{13\sigma^2/9}{11\sigma^2/9}$$

$$E_f = \frac{13\sigma^2}{9} \times \frac{9}{11\sigma^2} = \frac{13}{11}$$

13/11 is greater than 1.

Which shows that

$Var(T_1) < Var(T_2)$ thus T_2 is more efficient than T_1.

Question:15

Find value of x^2 test statistic $n=2.5$, $S^2=3$, $\sigma^2 = 18$.

Solution:

As we know that,

$$x^2 = \frac{nS^2}{\sigma^2}$$

putting values,

$$x^2 = \frac{(2.5)3}{18}$$

$$x^2 = 0.4167$$

Question:16

Write the variance of hypergeometric distribution.

Ans:

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$$Variance = \frac{nk}{N}\left(1-\frac{k}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Question:17

find proportion of X_1 and X_2 if $X_1 = 300$, $n_1 = 400$

$X_2 = 200$, $n_2 = 300$

Solution:

Proportion of X_1.

$$P_1 = \frac{X_1}{n_1} = \frac{300}{400} = \frac{3}{7}$$

$$\Rightarrow P_1 = 0.75 = 75\%$$

Proportion of X_2.

$$P_2 = \frac{X_2}{n_2} = \frac{200}{300} = \frac{2}{3}$$

$$\Rightarrow P_2 = 0.667 = 66.7\%$$

Question:18.

what is joint distribution?

Solution:

The distribution of two or more random variables which are observed simultaneously when an experiment is performed is called Joint distribution.

Question:19

$n = 1150$, $x = 450$, $P_0 = 0.39$

$H_0 : P_0 = 0.39$, $\alpha = 0.01$

Stated hypothesis.

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Solution:

1) ***Formulation:***

$H_0 = P_0 = 0.39$

$H_1 = P_0 \neq 0.39$

2) ***LOS***

$\alpha=0.01$, $\alpha/2=0.01/2=0.005$,

$1-\alpha/2=0.995$

3) ***Test statistic***

$$Z=\frac{X-np_0}{\sqrt{np_0q_0}}$$

$$Z=\frac{(x\pm 1/2)-np_0}{\sqrt{np_0q_0}}\ldots\ldots(A)$$

4) ***Critical region***

$$Z_{1-\alpha/2}=Z_{0.995}=2.576$$
$$|Z|_c<2.576$$

5) ***Calculation***

1st of all we find q_0 from the given values,

$q_0 = 1- p_0$, $q_0 = 1-0.39$,
$q_0 = 0.61$

now Putting the values in eq.(A)

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$$Z = \frac{(450 - 1/2) - 1150(0.39)}{\sqrt{(1150)(0.39)(0.61)}}$$

$$Z = \frac{4495 - 448.5}{16.540}$$

$$Z = \frac{1}{16.540} = 0.0604$$

$$\Rightarrow Z = 0.0604$$

6) ***Conclusion***

Since,

$|Z|_c < 2.576$

0.0604 < 2.576

It does not lies in rejection region. So we do not reject H_0 against H_1 .

Question:20

What is the relationship b/w level of significance and critical region?

Ans:

A critical region specifies a set of values of the test statistic for which null hypothesis is rejected.

The actual type I error that we get by using critical region it will be the level of significance.

Question:21

What is g(x) and h(y)?

Ans:

From the joint probability function for (x , y), we can obtain the individual probability functions of X and Y.

Such individual probability functions are called marginal probability function.

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The marginal probability function of X :

g(x)=p(X=x)

The marginal probability function of Y :

h(y)=p(Y=y)

Question:22

Define point estimation.

Ans:

The object of point estimation is to obtain a single number from the sample of unknown true value of population parameter.

Question:23

Write 3 properties of normal distribution.

Ans:

1) ***Continuous distribution:***

   The normal probability distribution is a Continuous distribution that ranges from $-\infty$ to $+\infty$.

   $R_x = \{x: -\infty < x < +\infty\}$

2) ***Total distribution:***

   The total area under the curve is unity.

   That is $P(-\infty < x < +\infty) = 1$

3) ***Symmetric distribution:***

   The normal probability distribution is a Symmetric distribution.

   - **Mean =median=mode**
   - $X_{0.75} - \mu = \mu - X_{0.25}$
   - $\mu_1 = \mu_3 = \mu_5 = ,\ldots\ldots\ldots, = 0$

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Question:24

Why we use pooled proportion instead of individual Proportion?

Ans:

We use the pooled proportion instead of individual Proportion, because the pooled proportion estimate the standard error and individual not.

Question:25

What is difference b/w random design and random block design?

Ans:

Complete Randomized design:

A complete randomized(CR) design , which is the simplest type of the basic designs , may be defined as a design in which treatments are assigned to experimental units completely at random , that is the randomization done without any restriction.

Randomized Complete block design:

A Randomized Complete block (RCB) design, may be defined as a design in which

- **I) Experimental material is divided into groups or blocks.**
- **II) Each block contains a complete set of treatments.**
- **III) The treatments are assigned at random to the experimental units within restrictions.**

Question:26

How we can control type I error and type II error?

Answer:

To control type I error we can set a simple lower significance level.

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While control type II error is impossible to control completely, the only possible chance to control the type II error by increasing sample size.

Question: 27

n=15, α = 95% then find $t_{\alpha/2\,;\,v}$?

Solution:

Given that

n =15 , α = 95% = 0.15

α/2 = 0.95/2 = 0.025

and,

v = n-1 = 15-1=14

so,

$t_{0.025\,;\,14} = 2.145$

Question:28:

If TSS = 272.25 , $\sum_i \sum_j X^2 = 4953$ ***. Find (C.F) correction factor.***

Solution:

We know that

$$TSS = \sum_i \sum_j X_{ij}^2 - C.F$$

Putting values,

272.25 = 4953 - C.F

C.F = 4953 - 272.25

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C.F = 4680.75

Question:29

what is relation between mean and variance in poison distribution.

Solution:

The mean and variance in the poison distribution are same i.e. μ.

Question:30

The department claim that the exceeds Rs = .2500 at the 0.05 level , then formulate null alternative hypothesis?

Solution:

Formulation

$H_0 : \mu < 2500$

$H_1 : \mu > 2500$

Question:31

when expected frequency in any category is less then 5 what we use?

Solution:

If an expected frequency in any category is less than 5 , then we use Yate correction of continuity.

$$x^2 = \sum_{i=1}^{2} \frac{(|0-e|-1/2)^2}{e}$$

It is applied in the normal approximation to the binomial distribution.

Question:32.

If n=400, x=275. Then find 95% confidence interval for population proportion.

Solution:

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We know that,

$$\hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}.\hat{q}}{n}}.........(1)$$

Now,

$$\hat{p} = \frac{x}{n} = \frac{275}{400} = 0.69$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.69 = 0.31$$

$$1 - \alpha = 0.95$$
$$\Rightarrow \alpha = 0.05 \;\;, \alpha/2 = 0.025$$

$$putting\ all\ values\ in\ eq.(1)$$

$$0.69 \pm Z_{0.025}\sqrt{\frac{(0.69)(0.31)}{400}}$$

$$0.69 \pm Z_{0.025}(0.023)$$
$$0.69 \pm (1.96)(0.023)$$
$$0.69 \pm 0.04508$$
$$\Rightarrow 0.69 + 0.04508 \;\;,\;\; 0.69 - 0.04508$$
$$\Rightarrow 0.73508 \;\;,\;\; 0.645$$
$$\Rightarrow (0.645, 0.73508)$$

<u>Question:33.</u>

Explain chi-square method.

<u>Answer:</u>

The chai-square distribution contains only one parameter, called the number of degree of freedom (d. f). where the term d. f represents the number of independent random variable that expresses the chai-square.

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$$x^2 = \sum_{i=1}^{2} \frac{(|0-e|-e)^2}{e}$$

Question:34.

Why the statistics is called science of decision making. Explain it?

Answer:

In this it is called science of decision making because it is enabling us to establish numerical benchmarks and monitor and evaluate the progress of our policy or program.

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